Exp: 12.

End to End communication at transport layer

a) Implement echo client server using TCP/UDP sockets.

b) Implement a chat program using socket programming.

TCP:

```
import socket
import threading

# --- SERVER ---
def start_server():
    server = Socket.Socket(Socket.AF_INET,
                           Socket.SOCK_STREAM)
    server.bind(('127.0.0.1', 5000))
    server.listen(1)
    print("[SERVER] Listening on 127.0.0.1:5000")
    conn, addr = server.accept()
    print(f"[SERVER] connected by {addr}")
While True:
    data = conn.recv(1024).decode()
    if not data or data.lower() == 'quit':
        print("[SERVER] connection closed by client.")
        break
    print(f"[SERVED] Connection closed by client.")
    Conn.Send(f"server received: {data}'.encode())
Conn.close()
server.close()
```

```
# -- CLIENT --
def Start_client()
    client = socket.socket(socket.AF_INET, Socket.
                           SOCK_STREAM)
    client.connect(('127.0.0.1', 5000))
    print("[CLIENT] Connected to server")
While True:
    msg = input("Enter message (or 'quit' to exit:")
    client.send(msg.encode())
    if msg.lower() == 'quit':
        break
    response = client.recv(1024).decode()
    print(f"[SERVER RESPONSE] {response}")
client.close()

# --- MAIN --
if __name__ == "__main__":
    Server_thread = threading.Thread(target =
                    Start_Server, daemon = True)
    Server_thread.Start()
    Start_client()
```

Output:

Listening on 127.0.0.1:5000 [CLIENT]
Enter msg: hello
[SERVER] Recieved: hello
[SERVER RESPONSE] Server Recieved: hello
Enter message (or 'quit or Exit'): quit
Connection closed by client

UDP

```
import socket
import threading

# -- SERVER --
def udp_server():
    s = socket.socket(socket.AF_INET, socket.
                      SOCK_DIAGRAM)

    s.bind(('127.0.0.1', 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
    While True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower() = 'quit':
            print(f"[SERVER] client {addr}
                        ended the connection")

            break
        print(f"[SERVER] Recieved from {addr}:{msg}")

        s.sendto(data, addr) #Echo back

    s.close()
    print("[SERVER] stopped")

# -- CLIENT --
def udp_client():
    c = socket.socket(socket.AF_INET, socket,
                      SOCK_DGRAM)
    server = ('127.0.0.1', 12000)
```

```
While True:
    msg = input("Enter message (or 'quit' to
                stop):")
    c.Sendto( msg.encode().Server )
    if msg.lower()=='quit':
        print("[CLIENT] closing connection...")
        break
    data, _ = c.recvfrom(1024)
    print(" Echo from server :", data.decode())
    c.close()
    print("[CLIENT] stopped.")
#__ MAIN __
if __name__ ="__main__":
    threading.Thread(target = udp_server,
        daemon = True).start() udp_client()
```

Output:

```
Enter message (or 'quit' to stop) : [SERVER]
Running on 127.0.0.1:12000] hi
[SERVER] Received from ('127.0.0.1', 49670): hi
Echo from Server : hi
Enter Message (or 'quit' to stop) : hello
[SERVER] Received from ['127.0.0.1', 49670]: how are you
Echo from server : how are you
Enter message ( or 'quit' to stop) : quit
[CLIENT] stopped [SERVER] stopped.
```

Result:

Thus, the end to end communication at Transport Layer is executed the code.